

British Offshore Wind & Advanced Titanium Programme

A preliminary business case for selective titanium use in a proposed 1.5 GW offshore wind development.

Prepared for: Department for Energy Security and Net Zero (DESNZ)

Proposed sponsor: TitanCore Wind Ltd. (proposed entity; not represented as an existing registered company)

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Status: Initial business case for feasibility, market engagement and competitive procurement

IMPORTANT STATUS NOTE

This document is an indicative proposal, not an executed contract, an official Government submission, a site-specific engineering design, a financing offer or a legal opinion. Numerical values are planning assumptions unless otherwise identified. Independent technical, commercial, legal and procurement review is required.

TITANCORE WIND

Selective advanced materials. Reliable renewable generation. British supply-chain ambition.

1. Executive Summary

TitanCore Wind proposes a British offshore wind programme combining utility-scale renewable electricity generation with selective use of titanium in components exposed to demanding corrosion, fatigue and operating conditions. The proposal is not to manufacture an entire turbine from titanium, but to assess where titanium can deliver measurable lifecycle benefits while retaining established materials for major structures.

The initial reference project comprises 100 turbines rated at 15 MW each, for an illustrative installed capacity of 1.5 GW. A net capacity factor of 46% gives an indicative annual generation of approximately 6.04 TWh. These are scenario assumptions, not a site-specific forecast.

Headline metric	Illustrative value	Basis
Installed capacity	1.5 GW	100 × 15 MW
Annual net generation	6.04 TWh	46% net capacity factor
Titanium requirement	500 tonnes	5 t per turbine, provisional allowance
Titanium material budget	£25 million	500 t × assumed £50,000/t
Five-year straight-line allocation	£5 million/year	Commercial costing illustration

The immediate request is for feasibility work and market engagement—not an unconditional public funding commitment or a guaranteed government-backed purchase.

2. Project Scope and Strategic Rationale

The programme has three proposed objectives: (1) develop a 1.5 GW offshore wind reference project; (2) assess opportunities for a British titanium-component supply chain; and (3) reduce whole-life operating costs where component testing demonstrates improved corrosion performance, component life, maintenance outcomes or weight.

The Government's Clean Energy Industries Sector Plan identifies offshore wind as an important UK industrial opportunity, including domestic supply-chain development. This proposal should be assessed against current energy, industrial and procurement policy before any formal submission.

The technical case for titanium must be demonstrated component by component. Titanium has lower density than steel and can offer strong corrosion resistance, but its elastic modulus is lower and its cost is materially higher. A substitution therefore requires assessment of geometry, fatigue, buckling, joining, manufacturing, inspection and lifecycle cost.

3. Proposed Material Architecture

Turbine system	Proposed approach	Potential titanium application
Blades	Composite structures	No general titanium replacement proposed
Tower	Predominantly structural steel	Specialist fittings only if justified
Fixed or floating support	Primarily qualified steel systems	Selected corrosion-critical interfaces, subject to validation
Fasteners and fittings	Load- and environment-specific materials	Potential high-value titanium use
Heat exchangers / fluid systems	Application-specific alloys	Titanium where corrosion resistance warrants it
Nacelle and drivetrain	Established structural/mechanical materials	Selected components after fatigue and cost validation
Electrical systems	Established conductors and components	No general substitution proposed

The proposal does not advocate replacing an entire turbine with titanium. Material choice must follow verified engineering requirements and a comparison against the best qualified conventional alternative.

4. Provisional Titanium Component Allocation

For preliminary costing only, assume an average of 5 tonnes of titanium-containing components per 15 MW turbine. This is an illustrative design-study allowance, not an established bill of materials.

Component category	Per turbine	Share
Fasteners, fittings and connection components	2.0 t	40%
Selected corrosion-critical fluid / heat-transfer components	1.5 t	30%
Specialist machined parts and interfaces	1.0 t	20%
Engineering allowance / other qualified components	0.5 t	10%
Total	5.0 t	100%

These percentages describe the proposed titanium allocation only, not the share of the entire turbine made from titanium. For 100 turbines, the assumed requirement is $100 \times 5 \text{ t} = 500 \text{ tonnes}$.

5. Wind-Generation Formula

The theoretical wind power intercepted by a rotor is:

$$P_{\text{wind}} = \frac{1}{2} \rho A v^3$$

Where P_{wind} is wind power (W), ρ is air density (kg/m^3), A is rotor swept area (m^2), and v is wind speed (m/s).

A simplified practical electrical-output equation is:

$$P_e = \min(P_{\text{rated}}, \frac{1}{2} \rho A v^3 C_p \eta)$$

Here C_p is the aerodynamic power coefficient and η represents drivetrain and electrical conversion efficiency. A real power curve must also reflect cut-in, rated and cut-out speeds, operating controls, wake effects and availability. The ideal Betz limit is approximately 59.3% of the power in the wind; actual extraction is lower.

Worked example: one 15 MW turbine

Parameter	Illustrative value
Rotor diameter	240 m
Air density	1.225 kg/m^3
Wind speed	10 m/s
Power coefficient, C_p	0.45
Drivetrain/electrical efficiency, η	0.94
Rated electrical power	15 MW

$$A = \pi D^2 / 4 = \pi \times 240^2 / 4 \approx 45,239 \text{ m}^2$$

Using the assumptions above gives an approximate electrical output of 11.7 MW at that wind speed. This is a simplified operating-point calculation, not a validated power curve or an independent prediction.

Titanium does not inherently increase the aerodynamic power coefficient. Any claimed increase in generation must be demonstrated using a matched baseline design and validated aerodynamic and operational models.

6. Annual Generation Estimate

For project-level energy estimation:

$$E_{\text{annual}} = P_{\text{installed}} \times 8,760 \times CF_{\text{net}}$$

Assume a 50% gross capacity factor and an 8% combined allowance for electrical, availability and other losses. Then $CF_{\text{net}} = 0.50 \times (1 - 0.08) = 0.46$.

$$E_{\text{annual}} = 1,500 \text{ MW} \times 8,760 \text{ h} \times 0.46 \approx 6.04 \text{ TWh/year}$$

Net capacity-factor scenario	Annual net generation
35%	4.60 TWh
40%	5.26 TWh
46%	6.04 TWh
50%	6.57 TWh

The capacity factor must ultimately come from site-specific wind-resource assessment, turbine power curves, wake modelling, grid constraints and a documented loss budget. Values above are scenario analysis only.

7. Titanium Procurement and Five-Year Allocation

The following model uses an assumed titanium material price of £50,000 per tonne. This is a planning assumption, not a current supplier quotation. Product form, grade, processing, origin, machining, scrap, certification and logistics can materially change the delivered cost.

Item	Calculation	Amount
Number of turbines	100	100
Titanium per turbine	Assumption	5 tonnes
Total titanium	$100 \times 5 \text{ t}$	500 tonnes
Assumed material price	Planning allowance	£50,000/t
Total material purchase value	$500 \times £50,000$	£25,000,000
Five-year straight-line allocation	$£25\text{m} \div 5$	£5,000,000/year
Allocation per turbine per year	$£50,000 \div 5$	£10,000/year

This five-year allocation is a commercial costing illustration, not a determination of statutory depreciation or the appropriate accounting useful life. Actual accounting treatment should follow the applicable policy and asset classification.

At 6.04 TWh per year, five-year generation would be approximately 30.2 million MWh. Dividing £25 million by this output gives about £0.83/MWh for the material allowance alone. This is not the project's levelised cost of electricity and excludes turbines, foundations, installation, grid connection, finance, operations, decommissioning and other capital costs.

8. Commercial Business Case

TitanCore Wind should proceed only if selected titanium components show a measurable lifecycle benefit against the best qualified conventional alternative. The potential value drivers are:

- **Corrosion-related maintenance:** fewer interventions or longer replacement intervals where evidence supports the claim.

- **Component life:** improvements demonstrated by relevant fatigue, corrosion and inspection data.
- **Weight reduction:** a verified reduction in the complete engineered assembly that creates a measurable cost or performance benefit.
- **British supply-chain development:** opportunities in processing, machining, inspection, qualification and component assembly.

Before investment approval, the financial model should include:

- Total installed capital expenditure and contingency.
- Titanium versus conventional-material cost differential.
- Annual operating and maintenance expenditure.
- Replacement and offshore intervention costs.
- Availability, net annual energy and revenue assumptions.
- Weighted average cost of capital, NPV and IRR.
- LCOE and sensitivities to titanium prices, interest rates, capacity factor and component failure rates.

9. Proposed Public-Private Procurement Structure

The proposal should not assume that a new PFI/PF2-style contract is available. The UK Government announced that it would no longer use PF2 for new government projects. A more appropriate concept is a competitively procured offshore-wind development with private project finance and an applicable electricity revenue arrangement, subject to legal and policy advice.

The Contracts for Difference scheme is a key low-carbon electricity support mechanism. Eligibility, allocation-round rules, standard terms and project-specific requirements must be checked against current official documentation before the project is structured.

Provision	Proposed position
Project company	Special-purpose project company to be established
Public counterparty	To be determined under the applicable statutory scheme and procurement
Private sponsor	TitanCore Wind consortium, subject to formation and qualification
Project scope	Finance, design, procure, construct, operate and maintain
Capacity	Illustrative 1.5 GW
Revenue mechanism	Competitive CfD or other legally available route, subject to eligibility
Titanium procurement	Competitive, specification-led, traceable procurement
Performance guarantees	Availability, power curve and component guarantees backed by evidence
Construction / operational risk	Allocated to parties best placed to manage each risk
Government payment	No unconditional payment guarantee proposed
Audit	Technical, financial, safety and supply-chain audit rights
Termination and disputes	Detailed default, cure, compensation and agreed dispute-resolution provisions

This is a proposed term sheet for discussion, not an executed contract, procurement award or legal opinion. The required seabed rights and development consents depend on the site and project characteristics and must be established with the relevant authorities.

10. Indicative Delivery Programme

Phase	Indicative timing	Activities
1. Feasibility and market engagement	0–12 months	Candidate components, baseline design, supplier quotations and site screening
2. Engineering qualification	12–24 months	Corrosion, fatigue, load, joining, manufacturing and lifecycle studies
3. Pilot deployment	24–48 months	Qualified components, controlled deployment, monitoring and baseline comparison
4. Commercial-scale procurement	Following evidence gates	Expand only where technical suitability, supply readiness and value for money are demonstrated

These are planning estimates, not guaranteed dates. They exclude any guaranteed planning, seabed leasing, grid-connection or construction timetable.

11. Key Risks and Mitigations

Risk	Mitigation
Titanium costs exceed budget	Competitive tenders, alternative grades and component-level selection
No measurable generation benefit	Do not attribute additional power to titanium without validated evidence
Corrosion or fatigue underperforms	Independent testing, traceability and monitored pilot deployment
Supply-chain dependence	Multiple qualified suppliers and documented sourcing
Planning, consent or grid delay	Early engagement with authorities and network stakeholders
Project cost escalation	Independent cost review, contingency and stage-gate investment decisions
Unproven lifecycle savings	Measured maintenance and failure data against conventional baseline
Procurement or subsidy challenge	Independent legal, procurement and subsidy-control review

12. Government Decision Requested

TitanCore Wind invites the UK Government to consider the following next steps:

- Commission or support an initial technical and commercial feasibility study.
- Undertake market engagement with offshore wind developers, turbine manufacturers, titanium suppliers and engineering institutions.
- Identify a competitive pilot opportunity for qualified corrosion-critical components.
- Assess whether verified domestic supply-chain investment could fit relevant industrial-policy and clean-energy support frameworks.
- Progress to full project development only after independent technical and financial validation.

The immediate request is to develop the evidence base and engage with the market—not to seek an unconditional £25 million titanium purchase commitment or guaranteed Government-backed project.

13. Conclusion

The opportunity is to make titanium a carefully selected enabling material within a British offshore wind supply chain, not to claim that titanium itself generates more wind power. Under the illustrative assumptions in this proposal, a 1.5 GW reference project could produce around 6.04 TWh net annually. A provisional 500-tonne titanium component programme would imply £25 million in material procurement at the assumed price, or £5 million per year when allocated evenly over five years.

The proposal becomes investable when a qualified design, site-specific energy assessment, supplier quotations, lifecycle-cost comparison and legally sound revenue and procurement structure substantiate the assumptions.

Recommended next step: commission an independent feasibility study comparing conventional components with selective titanium alternatives before seeking full Government procurement or project-finance approval.

Sources and Reference Notes

- UK Government, announcement on not using PF2 for new government projects (2018): <https://www.gov.uk/government/news/government-to-no-longer-use-pf2-for-new-projects>
- UK Government, Clean Energy Industries Sector Plan: <https://www.gov.uk/government/publications/clean-energy-industries-sector-plan>
- UK Government, Contracts for Difference Allocation Round 8 materials and standard terms: <https://www.gov.uk/government/collections/contracts-for-difference>
- UK Government, offshore wind planning and consenting guidance: <https://www.gov.uk/guidance/offshore-wind-farms-planning-and-consenting>
- UK Government, offshore wind cost and technical assumptions (check latest release): <https://www.gov.uk/government/collections/energy-generation-cost-projections>
- Theoretical wind power is commonly modelled using $P = \frac{1}{2}\rho A v^3$, with the ideal Betz limit of approximately 59.3%; actual turbine output depends on the power curve, site and losses.

All project quantities, turbine ratings, material proportions, titanium pricing, capacity factors and cost allocations in this document are illustrative planning assumptions unless explicitly identified otherwise. Verify all source links, current policies, market prices, technical specifications and regulatory requirements before formal submission.